

02425 Diffusions and Stochastic Differential Equations

Exercises Week 3: Stochastic Processes

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Question 1, simulation of Brownian motion Implement a function which simulates Brownian motion on an interval $[0, T]$; for example as given in the book: the function should take as input a partition $0 \leq t_1 < \dots < t_n = T$, and should compute and return $B_{t_1}, B_{t_2}, \dots, B_{t_{n-1}}, B_T$. Test the function by simulating sufficiently many replicates of $(B_0, B_{1/2}, B_{3/2}, B_2)$ to verify the covariance of this vector, and the distribution of B_2 . Save the function for future use.

We implement a function that takes a partition of time points distributed as given. That is, the partition *does not* necessarily follow a uniform distribution between the n points across T , unless specified. For this Δt is calculated for every time point t and corresponding ΔB is generated. We then verify the variance of B_2 as 2 with a mean 0 over 10_000 simulations.

Question 2, Extrema of Brownian motion and hitting times

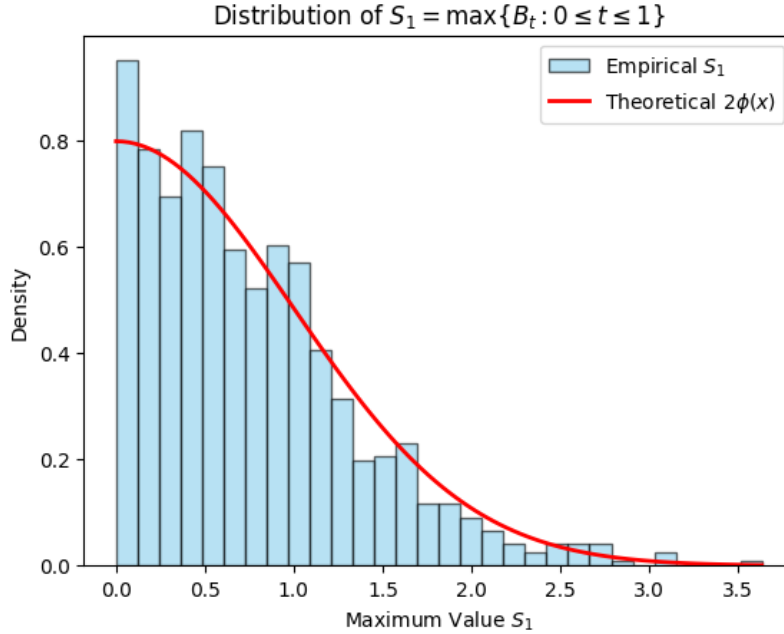
1. Generate $N = 1_000$ sample paths of Brownian motion of the time interval $[0, 1]$ using a time step of 0.001.

See `exercises.w3.ipynb` for solution.

2. For each sample path $\{B_t : 0 \leq t \leq 1\}$, compute the maximum $S_1 = \max\{B_t : 0 \leq t \leq 1\}$. Plot the histogram of S_1 (or the empirical c.d.f.) and compare with the theoretical distribution of S_1 .

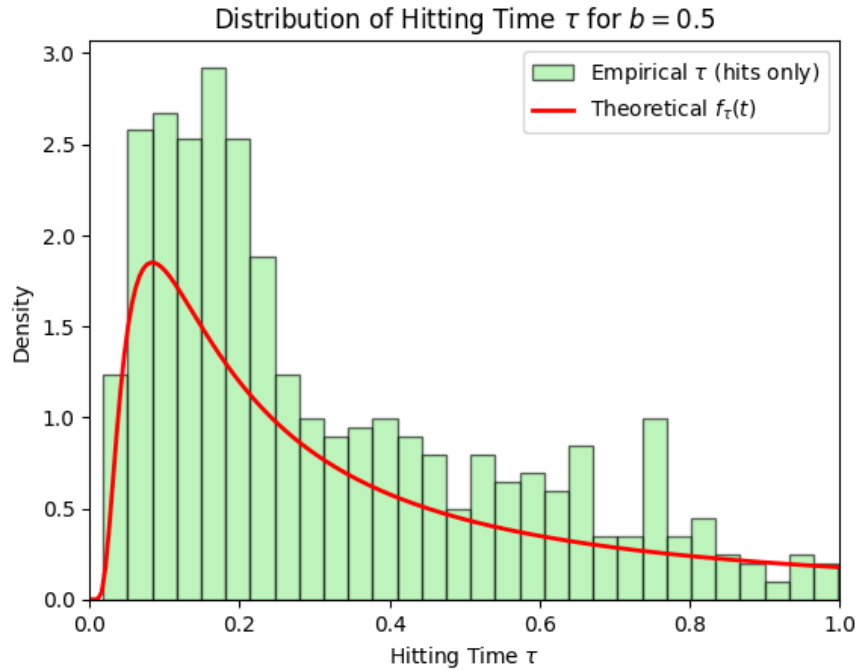
The theoretical distribution of S_1 is defined by the reflection principle such that $P(S_1 \geq x) = 2P(B_1 \geq x)$, where $P(B_1 \geq x) = 2\Phi(-x\sqrt{1}) = 2(1 - \Phi(x))$ since B_1 is normally distributed. The derivative of this is simply $f_{S_1}(x) = 2\phi(x)$ for $x \geq 0$. *Note:* Φ is the cumulative distribution function (cdf) of a normal distribution, i.e. the total accumulated probability from $-\infty$ to x (area under the bell curve up to x). ϕ is the probability density function (pdf) of a normal distribution, defined as $\phi = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. It is the exact height of the bell curve at x .

We plot the generated paths against this theoretical distribution;



3. For each sample path $\{B_t : 0 \leq t \leq 1\}$, compute the hitting time $\tau = \min\{t : B_t \geq b\}$ with $b = 0.5$. *Note:* if sample path does not hit b in the time interval $[0, 1]$, then define $\tau = 1$. Plot the histogram of τ (or the empirical c.d.f.) and compare with the theoretical distribution.

We filter all paths that hit b within the interval, and plot the hitting time τ in a histogram against the theoretical distribution. *Note:* the theoretical distribution is given as $f_\tau(t) = bt^{-3/2}\phi(b/\sqrt{t})$ by theorem 4.3.3 in the book.



Question 3, total and quadratic variation of Brownian motion Reproduce figure 4.3 (page 70) in the book:

1. Generate one sample path of standard Brownian motion on the time interval $[0, 1]$ using a time step of 2^{-20} .

See `exercises.w3.ipynb` for solution.

2. Compute the discretized total variation $V_\Delta = \sum |\Delta B|$ and quadratic variation $|B|_1 = \sum |\Delta B|^2$.

See `exercises.w3.ipynb` for solution.

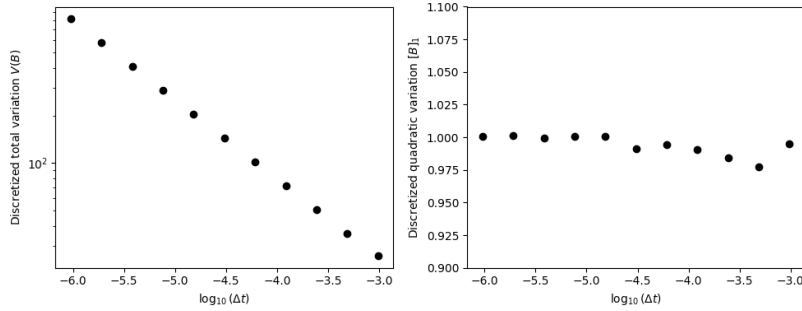
3. Subsample the Brownian motion at every other time step.

See `exercises.w3.ipynb` for solution.

4. Repeat the previous steps until you have reached a time step of 2^{-10} .

See `exercises.w3.ipynb` for solution.

5. Plot the discretized total variation and quadratic variation as function of the time step in double logarithmic plot.



Include the analytical predictions in the graphs, using exercise 4.20 (page 88) for the total variation. *Optional:* solve the exercise 4.20 so that you know where the analytical prediction comes from.

Exercise 4.20 Let $\Delta = \{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}$ be a partition of $[0, 1]$, where $n \in \mathbb{N}$, and let V_Δ be the discretized total variation, $V_\Delta = \sum_{i=1}^{n-1} |B_{t_i} - B_{t_{i-1}}|$.

1. Show that $\mathbb{E}V_\Delta \sim \sqrt{n}$, (in fact $\mathbb{E}V_\Delta = \sqrt{2n/\pi}$ so that $\mathbb{E}V_\Delta \rightarrow \infty$ as $n \rightarrow \infty$).

For a variable $X \sim \mathcal{N}(0, \sigma^2)$, the expected absolute value $\mathbb{E}|X|$ is given by the PDF of the normal distribution

$$\mathbb{E}|X| = \int_{-\infty}^{\infty} |x| \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx$$

which simplifies to $\mathbb{E}|X| = \sigma\sqrt{\frac{2}{\pi}}$, and since our specific variance $\sigma^2 = \frac{1}{n}$ (from Δ it is given that $\Delta t = \frac{1}{n}$), we have that

$$\mathbb{E}|\Delta B_i| = \sqrt{\frac{2}{n\pi}}$$

Now, because V_Δ contains n identical expectations, we multiply the result by n ;

$$\mathbb{E}V_\Delta = \sum_{i=1}^n \mathbb{E}|\Delta B_i| = n\sqrt{\frac{2}{n\pi}} = \sqrt{\frac{2n}{\pi}}$$

Now we can see that as $n \rightarrow \infty$ the term $\sqrt{\frac{2n}{\pi}}$ diverges to ∞ .

2. Show that $V_\Delta \rightarrow \infty$ w.p. 1 as $n \rightarrow \infty$, using

$$\sum_{i=1}^n |B_{t_i} - B_{t_{i-1}}|^2 \leq \max_i \{|B_{t_i} - B_{t_{i-1}}|\} \cdot \sum_{i=1}^n |B_{t_i} - B_{t_{i-1}}|$$

as well as the quadratic variation and continuity of Brownian motion.

The left term is the discretized quadratic variation, and the right term is the biggest time step multiplied with the total variation. We know that the quadratic variation of Brownian motion converges to the total time elapsed, which for the interval $[0, 1]$ is 1. And for the term $\max_i \{|\Delta B_i|\}$, it must shrink towards 0 as $\Delta t = \frac{1}{n} \rightarrow 0$ (when $n \rightarrow \infty$). So the inequality given now is

$$1 \leq 0 \cdot \sum_{i=1}^n |\Delta B_i|$$

If this is to hold, then the total variation (last part of the right term, which is the same as V_Δ) must diverge to ∞ as $n \rightarrow \infty$. If it diverged to some finite value, the right side would be 0, and then the inequality would not hold.

Basic Martingales

Question 4 Show that $\{B_t^2 - t\}$ is a martingale (exercise 4.8 in the book; page 79); Let $\{B_t\}$ be Brownian motion w.r.t. a filtration $\{\mathcal{F}_t\}$. Show that the process $\{B_t^2 - t : t \geq 0\}$ is a martingale w.r.t. $\{\mathcal{F}_t\}$. *Note:* the so-called Lévy characterization of Brownian motion says that a converse result also holds: if $\{M_t\}$ is a continuous martingale such that also $\{M_t^2 - t\}$ is a martingale, then $\{M_t\}$ is Brownian motion.

A martingale must satisfy

1. Transformation adapts to filtration; when B_t is adapted to $\mathcal{F}_s$ then $B_t^2 - t$ is also adapted to $\mathcal{F}_s$.
2. The expected absolute value should be finite; $\mathbb{E}[|M_t|] < \infty$
3. The martingale property; $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$, $\forall t > s \geq 0$.

1) is given. For 2) we have that $B_t \sim \mathcal{N}(0, t)$ such that its variance is $\mathbb{E}[B_t^2] = t$. Thereby, $\mathbb{E}[|B_t^2 - t|]$ is finite. For 3) we substitute $B_t = B_s + (B_t - B_s)$ to get $\mathbb{E}[(B_s + (B_t - B_s))^2 | \mathcal{F}_s]$ we expand the square

$$\mathbb{E}[B_s^2 + 2B_s(B_t - B_s) + (B_t - B_s)^2 - t | \mathcal{F}_s]$$

Now, since $\mathbb{E}$ is a linear operator we take one term at a time;

$$\begin{aligned} \mathbb{E}[B_s^2 | \mathcal{F}_s] &= B_s^2 \\ \mathbb{E}[2B_s(B_t - B_s) | \mathcal{F}_s] &= 0 \\ \mathbb{E}[(B_t - B_s)^2 | \mathcal{F}_s] &= t - s \end{aligned}$$

Note: second term carries $(B_t - B_s)$ which is the future increment, which has the mean 0. For any variable, the expected value of its square is equal to its variance. And since the Brownian increment is normally distributed with variance $t - s$ it evaluates to this.

If we substitute back in to our expression we get

$$\mathbb{E}[B_s^2 + 2B_s(B_t - B_s) + (B_t - B_s)^2 - t \mid \mathcal{F}_s] = B_s^2 + 0 + (t - s) - t = B_s^2 - s$$

Thus, proving that $\mathbb{E}[B_t^2 - t \mid \mathcal{F}_t] = B_s^2 - s$. $\{B_t^2 - t\}$ fulfils the martingale property.

Question 5 Solve exercise 4.19 in the book (page 87) concerning Doob's martingale; Let X be a random variable such that $\mathbb{E}|X| < \infty$, and let $\{\mathcal{F}_t : t \geq 0\}$ be a filtration. Show that the process $\{M_t : t \geq 0\}$ given by $M_t = \mathbb{E}\{X \mid \mathcal{F}_t\}$ is a martingale w.r.t. $\{\mathcal{F}_t : t \geq 0\}$.

Again, a martingale must satisfy the above mentioned 3 points. 1) again is given. 2) is shown by the standard property on conditional expectation $|\mathbb{E}[X \mid \mathcal{F}_t]| \leq \mathbb{E}[|X| \mid \mathcal{F}_t]$, and since $M_t = \mathbb{E}[X \mid \mathcal{F}_t]$ we can write $|M_t|$ in the left term; $|M_t| \leq \mathbb{E}[|X| \mid \mathcal{F}_t]$. Then take the full expectation on both sides, $\mathbb{E}[|M_t|] \leq \mathbb{E}[\mathbb{E}[|X| \mid \mathcal{F}_t]]$ which by the tower property simplifies to $\mathbb{E}[|M_t|] \leq \mathbb{E}[|X|]$. Now, since $\mathbb{E}[|X|] < \infty$ is given we have the finite expectation guaranteed. 3) we substitute the term for M_t into the left term of the martingale property; $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s]$. Since $\mathcal{F}_t$ is a larger information set than $\mathcal{F}_s$, the smaller σ -algebra dictates the final result;

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s]$$

This is exactly the expression we are given for M_s . Therefore, $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$.

Question 6 Solve exercise 4.10 (page 82) concerning the increasing variance of a martingale; Show that if $\{M_t : t \geq 0\}$ is a martingale such that $\mathbb{E}|M_t|^2 < \infty$ for all t , then the increments

$$M_t - M_s \text{ and } M_v - M_u$$

are uncorrelated, whenever $0 \leq s \leq t \leq u \leq v$. (*Hint:* when computing the covariance $\mathbb{E}(M_v - M_u)(M_t - M_s)$, condition on $\mathcal{F}_u$) Next, show that the variance of increments is additive:

$$\mathbb{V}(M_u - M_s) = \mathbb{V}(M_u - M_t) + \mathbb{V}(M_t - M_s)$$

Finally, show that the variance is increasing, i.e.

$$\mathbb{V}(M_s) \leq \mathbb{V}(M_t) \text{ whenever } 0 \leq s \leq t$$

First, to show that the two increments are uncorrelated we show that their covariance is 0. Covariance is defined as $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. But because the expected value of any martingale property is 0 we write

$$\mathbb{E}[(M_t - M_s)(M_v - M_u)]$$

We follow the hint provided and construct the conditional expected value of the covariance on $\mathcal{F}_u$

$$\mathbb{E}[\mathbb{E}[(M_t - M_s)(M_v - M_u)] \mid \mathcal{F}_u]$$

Since $0 \leq s \leq t \leq u \leq v$ we know the exact value of events $\leq u$, thus $\mathbb{E}[(M_t - M_s) \mid \mathcal{F}_u] = (M_t - M_s)$ such that we can pull it out of the inner conditional expectation;

$$(M_t - M_s) \cdot \mathbb{E}[(M_v - M_u) \mid \mathcal{F}_u]$$

Which we can further divide as

$$(M_t - M_s) \cdot \underbrace{(\mathbb{E}[M_v \mid \mathcal{F}_u])}_{(1)} - \underbrace{(\mathbb{E}[M_u \mid \mathcal{F}_u])}_{(2)}$$

(1): for a martingale property M_v , the expected future position given $\mathcal{F}_u$ is M_u
(2): M_u given $\mathcal{F}_u$ is also M_u
So the expression simplifies to

$$(M_t - M_s) \cdot (M_u - M_u) = (M_t - M_s) \cdot 0 = 0$$

Thus, the covariance is 0.

To prove additivity of the two increments we write them as

$$(M_u - M_s) = (M_u - M_t) + (M_t - M_s)$$

For two uncorrelated variables, the sum of their variance is equal to the variance of their sum;

$$\mathbb{V}[M_u - M_s] = \mathbb{V}[(M_u - M_t) + (M_t - M_s)] = \mathbb{V}[M_u - M_t] + \mathbb{V}[M_t - M_s]$$

Since we proved that the two increments are uncorrelated in the first part of this proof, we have now proven the additive property.

Finally, to show that the variance is increasing, we apply the variance operator to the expression for a future position M_t as the sum of the past position M_s and the increment;

$$\mathbb{V}[M_t] = \mathbb{V}[M_s + (M_t - M_s)]$$

And since M_s and the future increment $(M_t - M_s)$ are uncorrelated, we can write

$$\mathbb{V}[M_t] = \mathbb{V}[M_s] + \mathbb{V}[M_t - M_s]$$

Now, since the variance of any random variable is non-negative, we know that $\mathbb{V}[M_t - M_s] \geq 0$. And substituting this back into the expression by dropping the variance of the increment then gives us

$$\mathbb{V}[M_t] \geq \mathbb{V}[M_s]$$

Thereby proving that the variance is increasing.